

# 40.520 Stochastic Models

## Martingale Theory

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Spring 2026

# Outline

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1. Review on Conditional Expectation
2. Martingale Definition
3. Martingale Convergence Theorem
4. Martingale Stopping Theorem
5. Martingale Inequalities

## Conditional Expectation

## Conditional Expectation

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- Given two events  $A$  and  $B$  with  $P(A) > 0$ , define the conditional probability  $P(B|A) = \frac{P(A \cap B)}{P(A)}$ .
  - If  $P(A) = 0$ , the usual convention is that  $P(B|A) = 0$ .
- Given a discrete random variable  $Y$ , we define its conditional expectation with respect to the event  $A$  by

$$E[Y|A] = \sum_y y \cdot P(Y = y|A)$$

## Conditional Expectation

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$$f(x) = E[Y|X=x] = \sum_y y \cdot P(Y=y|X=x)$$

- When the event  $A = \{X = x\}$  where  $X$  is another discrete random variable, we define the function

$$f(x) = E[Y|X = x]$$

- We define  $E[Y|X] = f(X)$ , which means that  $E[Y|X]$  is itself a random variable.

# Properties

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- If  $X \perp Y$ , then  $E[Y|X] = E[Y]$
- If  $Y = g(X)$ , then  $E[YZ|X] = YE[Z|X]$
- Tower Property:

$$E[E[X|Y, Z]|Y] = E[E[X|Y]|Y, Z] = E[X|Y]$$

and in particular,  $E[E[Y|X]] = E[Y]$

- $E[C|Y] = C$  for any constant number  $C$

## Proof: Tower Property

Want to show that  $E[E[X|Y, Z]|Y] = E[X|Y]$

$$f(y, z) = E[X|Y=y, Z=z] = \sum_x x \cdot \frac{P(X, Y, Z)}{P(Y, Z)}$$

$$\begin{aligned}\therefore E[E[X|Y=y, Z=z]|Y=y] &= E[f(y, z)|Y=y] \\ &= \sum_z \left( \sum_x x \cdot \frac{P(X, Y, Z)}{P(Y, Z)} \right) \cdot \frac{P(Y, Z)}{P(Y)}\end{aligned}$$

$\downarrow$   
 $f(y, z)$

$$= \sum_z \sum_x x \cdot \frac{P(X, Y, Z)}{P(Y)}$$

$$\begin{aligned}&= \sum_x x \cdot \frac{\sum_z P(X, Y, Z)}{P(Y)} = \sum_x x \cdot \frac{P(X, Y)}{P(Y)} \\ &= E[X|Y=y] \quad \# \end{aligned}$$

Want to show that  $E[E[X|Y]|Y, Z] = E[X|Y] = f(Y)$

$$f(y) = E[X|Y=y] = \sum_x x \cdot \frac{P(X, Y)}{P(Y)} = \text{constant}$$



## Definition of Martingale

## What is a Martingale?

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A stochastic process  $\{Z_n, n = 1, 2, \dots\}$  is said to be a martingale process with respect to  $\{X_1, X_2, \dots\}$  if

$$E[Z_n | X_1, \dots, X_{n-1}] = Z_{n-1} \text{ for all } n.$$

A martingale can be viewed as a generalized version of a fair game. We have

$$E[Z_n] = E[Z_{n-1}].$$

**Key insight:** The best forecast for tomorrow is what we know today — no drift!

## Example: A Fair Play

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- A gambler plays a game repeatedly. At each play he wins \$1 with prob. 50% and loses \$1 with prob. 50%. The plays are independent of each other.
- Let  $X_n$  be the outcome of each play and

$$Z_n = \sum_{i=1}^n X_i$$

Show  $Z_n, n \geq 1$  is a martingale.

$$E[Z_{n+1} | X_1, \dots, X_n] = Z_n \quad ??$$

$$\begin{aligned} E[Z_{n+1} | X_1, \dots, X_n] &= E\left[\sum_{i=1}^{n+1} X_i \mid X_1, \dots, X_n\right] \\ &= E\left[\sum_{i=1}^n X_i + X_{n+1} \mid X_1, \dots, X_n\right] \\ &= \sum_{i=1}^n X_i + E[X_{n+1} | X_1, \dots, X_n] \\ &= Z_n + E[X_{n+1}] \\ &= Z_n \neq \end{aligned}$$

## Example: Product of Random Numbers

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- Let  $\{X_1, X_2, \dots\}$  be a sequence of independent random variables with

$$E[X_i] = 1$$

- Show

$$Z_n = \prod_{i=1}^n X_i, \quad n \geq 1,$$

is a martingale.

Want to show that  $E(Z_{n+1} | x_1, \dots, x_n) = Z_n$ :

$$\begin{aligned} E(Z_{n+1} | x_1, \dots, x_n) &= E\left[\prod_{i=1}^{n+1} X_i | x_1, \dots, x_n\right] \\ &= E\left[X_{n+1} \cdot \prod_{i=1}^n X_i | x_1, \dots, x_n\right] \\ &= \prod_{i=1}^n X_i \cdot E[X_{n+1} | x_1, \dots, x_n] \\ &= Z_n \cdot E[X_{n+1}] \\ &= Z_n \cdot (1) \\ &= Z_n \quad \# \end{aligned}$$

## Example: Variance of a Sum as a Martingale

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- Let  $\{X_1, X_2, \dots\}$  be a sequence of independent and identically distributed random variables with

$$E[X_i] = 0 \quad \text{and} \quad \text{Var}[X_i] = \sigma^2$$

- Let

$$Y_n = \left( \sum_{i=1}^n X_i \right)^2 - n\sigma^2$$

- Show  $\{Y_n, n = 1, 2, 3, \dots\}$  is a martingale.

$$\begin{aligned}
 E[X_{n+1} | x_1, \dots, x_n] &= E\left[\left(\sum_{i=1}^{n+1} x_i\right)^2 - (n+1)6^2 \mid x_1, \dots, x_n\right] \\
 &= E\left[\left(\sum_{i=1}^n x_i\right)^2 + 2\left(\sum_{i=1}^n x_i\right)x_{n+1} + x_{n+1}^2 \mid x_1, \dots, x_n\right] - (n+1)6^2
 \end{aligned}$$

$$\begin{aligned}
 Y_n &= \left[\left(\sum_{i=1}^n x_i\right)^2 - n6^2\right] + 2\left(\sum_{i=1}^n x_i\right)E[X_{n+1} | x_1, \dots, x_n] \\
 &\quad + \left[E[X_{n+1}^2 | x_1, \dots, x_n] - 6^2\right]
 \end{aligned}$$

$$\begin{aligned}
 &\quad \text{Var}[X_{n+1}] + (E[X_{n+1}])^2 = 6^2 + 0^2 = 6^2 \\
 &= Y_n + 0 + [6^2 - 6^2] \\
 &= Y_n \neq
 \end{aligned}$$



**Example: Pólya's Urn**  $\rightarrow E[Z_{n+1}] = E[Z_n]$  to show that it's  
Martingale process  $\square$

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Start with  $r$  red balls,  $b$  blue balls. At each step:

- Draw a ball uniformly
- Return it + add another ball of the same color

Let  $R_n$  = number of red balls after  $n$  steps. Let  $Z_n = \frac{R_n}{n + r + b}$  = proportion of red balls.

**Result: The ratio does not explode toward the color with more initial balls.**

$$\begin{aligned}
E[Z_{n+1} | R_1, \dots, R_n] &= E\left[\frac{R_{n+1}}{n+1+r+b} \mid R_1, \dots, R_n\right] \\
&= E\left[\frac{R_n + \frac{R_n}{n+r+b}}{n+1+r+b} \mid R_1, \dots, R_n\right] \\
&= \frac{n+1+r+b}{n+r+b} \cdot R_n \\
&= \frac{R_n}{n+r+b} \\
&= Z_n \neq
\end{aligned}$$

$$\therefore E[Z_{n+1}] = E[Z_n] = \dots = E[Z_0] = \frac{r}{r+b}$$

# Martingale Convergence Theorem

# Martingale Convergence

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- Does a martingale converge?
- Will  $Z_n$  approach a limit as  $n \rightarrow \infty$ ?

**Not all martingales converge** — we need an extra condition.

# Martingale Convergence Theorem

## Theorem (Martingale Convergence Theorem)

Let  $(Z_n)_{n \geq 1}$  be a martingale. If

$$\sup_n \mathbb{E}[|Z_n|] < \infty$$

then:

1. There exists a random variable  $Z_\infty$  with  $\mathbb{E}[|Z_\infty|] < \infty$
2.  $Z_n \rightarrow Z_\infty$  **almost surely**

**Key Insight:** If the expected absolute value doesn't blow up, the martingale must settle down.

## Example: Symmetric 1D Random Walk

$$X_i = \begin{cases} 1 & ; p = \frac{1}{2} \\ -1 & ; p = \frac{1}{2} \end{cases}$$

$$\text{CLT: } \frac{Z_n}{\sqrt{n}} \sim N(0,1) \text{ as } n \rightarrow \infty$$
$$Y \sim N(0,1)$$

Let  $Z_n = \sum_{i=1}^n X_i$  with  $P(X_i = \pm 1) = 1/2$

- Is  $Z_n$  a martingale? ✓
- Does it converge? ✗

$$E(|Y|) = \int_{-\infty}^{\infty} |y| \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

$$= 2 \int_0^{\infty} y \left( \frac{1}{\sqrt{2\pi}} \right) e^{-\frac{y^2}{2}} dy = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-u} du = \sqrt{\frac{2}{\pi}}$$

$$\text{Let } u = \frac{y^2}{2} \rightarrow du = y dy$$

$$\therefore \text{Var}[X_i] = 1$$

$$\therefore E\left[\left|\frac{Z_n}{\sqrt{n}}\right|\right] = \sqrt{\frac{2}{\pi}} \text{ as } n \rightarrow \infty$$

$$\therefore E[|Z_n|] = \sqrt{\frac{2n}{\pi}} \rightarrow \infty$$

$\therefore Z_n$  doesn't converge #

## Example Gambler's Ruin

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A gambler starts with \$ $k$ , aims for \$ $N$ , or goes broke. Let  $Z_n$  = wealth at time  $n$ .

- Is  $Z_n$  a martingale?  $\rightarrow$  Not always if  $p \neq q$ .
- Does it converge? What are the possible limits?

$$Z_n = \sum_{i=1}^n X_i, \text{ where } X_i = \begin{cases} 1 & \text{w.p. } p \\ -1 & \text{w.p. } 1-p = q \end{cases}$$

$$E[Z_{n+1} | X_1, \dots, X_n] = Z_n + E[X_{n+1}]$$

$$= Z_n + p - q \rightarrow \text{Not martingale if } p \neq q \\ \text{(unfair game)}$$

Consider fair game case:

$$Z_\infty = \begin{cases} 0 & \text{w.p. } 1-u(k) \\ N & \text{w.p. } u(k) \end{cases}$$

Define  $u(k) = P(\text{reach } N \text{ before } 0, \text{ starting from } k)$ ,  $u(0) = 0$ ,  
 $u(N) = 1$ . For  $k \in [1, N-1]$ ,

$$u(k) = \frac{1}{2} \cdot u(k+1) + \frac{1}{2} \cdot u(k-1).$$



## Martingale Stopping Theorem

# Stopping Times

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- The positive integer-valued (possibly infinite) random variable  $N$  is said to be a stopping time for a process  $\{Z_n, n = 1, 2, \dots\}$  if the occurrence of event  $\{N \leq n\}$  is determined by the value of  $\{Z_1, Z_2, \dots, Z_n\}$ .

## Example: Hitting Time of a Process

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Consider a process  $\{Z_n, n = 1, 2, \dots\}$ . Let

$$N = \min\{n : Z_n \geq 1\},$$

Is  $N$  a stopping time?



## Example: Maximum Time

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Consider a process  $\{Z_n, n = 1, 2, \dots\}$ . Let

$$N = \min \left\{ n : Z_n = \max_{1 \leq i \leq 10} Z_i \right\}$$

i.e.,  $N$  is the first time that the process reaches the largest value of  $Z_1, Z_2, \dots, Z_{10}$ .

Is  $N$  a stopping time? ~~X~~

Lo If  $\{N=7\}$ , it depends on  $Z_8, Z_9, Z_{10}$  also.

# Martingale Stopping Theorem

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## Theorem (Martingale Stopping Theorem)

If  $\{Z_n, n = 1, 2, \dots\}$  is a martingale with respect to  $\{X_1, X_2, \dots\}$  and  $T$  is a stopping time, then

$$E[Z_{T \wedge n}] = E[Z_1]$$

where  $T \wedge n = \min(T, n)$ .

$$\begin{aligned}
 E[z_{T \wedge n}] &= E\left[\sum_{j=1}^n z_{T \wedge n} \cdot \mathbb{1}_{\{T=j\}} + z_{T \wedge n} \cdot \mathbb{1}_{\{T>n\}}\right] \\
 &= \sum_{j=1}^n E[z_j \cdot \mathbb{1}_{\{T=j\}}] + z_n \cdot \mathbb{1}_{\{T>n\}}
 \end{aligned}$$

$$\begin{aligned}
 E[z_j \cdot \mathbb{1}_{\{T>j\}}] &= E[z_j \cdot \mathbb{1}_{\{T>j-1\}} - z_j \cdot \mathbb{1}_{\{T>j\}}] \\
 &= E[E[z_j \cdot \mathbb{1}_{\{T>j-1\}} | z_1, \dots, z_{j-1}]] - E[z_j \cdot \mathbb{1}_{\{T>j\}}] \\
 &= E[\mathbb{1}_{\{T>j-1\}} E[z_j | z_1, \dots, z_{j-1}]] - E[z_j \cdot \mathbb{1}_{\{T>j\}}] \\
 &= E[z_{j-1} \cdot \mathbb{1}_{\{T>j-1\}}] - E[z_j \cdot \mathbb{1}_{\{T>j\}}]
 \end{aligned}$$

Def. of Martingale

$$\begin{aligned}
 \therefore E[z_{T \wedge n}] &= E[z_1 \cdot \mathbb{1}_{\{T=1\}}] + \sum_{j=2}^n (E[z_{j-1} \cdot \mathbb{1}_{\{T>j-1\}}] - E[z_j \cdot \mathbb{1}_{\{T>j\}}]) + z_n \cdot \mathbb{1}_{\{T>n\}} \\
 &= E[z_1 \cdot \mathbb{1}_{\{T=1\}} + z_1 \cdot \mathbb{1}_{\{T>1\}}] \\
 &= E[z_1 \cdot \mathbb{1}_{\{T \geq 1\}}] = E[z_1] \neq
 \end{aligned}$$

# Martingale Stopping Theorem

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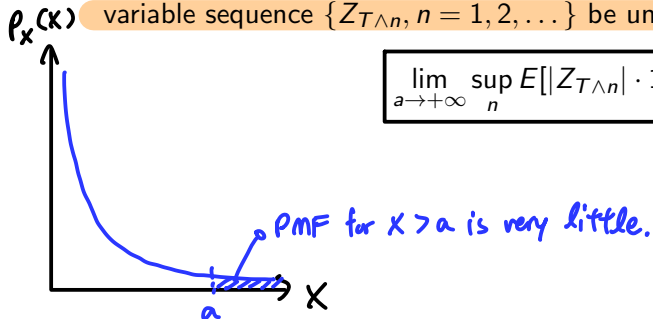
- Whenever we can justify the interchange of limit and expectation,

$$E[Z_1] = \lim_{n \rightarrow +\infty} E[Z_{T \wedge n}] = E \left[ \lim_{n \rightarrow \infty} Z_{T \wedge n} \right] = E[Z_T].$$

## Technical Conditions for the Interchange

- We return to the matter of justifying an interchange of limit and expectation as  $n \rightarrow +\infty$ .
- The most general conditions that guarantee this operation are that the random variable sequence  $\{Z_{T \wedge n}, n = 1, 2, \dots\}$  be uniformly integrable in the sense that

$$\lim_{a \rightarrow +\infty} \sup_n E[|Z_{T \wedge n}| \cdot 1_{\{|Z_{T \wedge n}| \geq a\}}] = 0.$$





## Technical Conditions for Interchange

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The uniform integrability is typically difficult to verify. But we do have several simpler sufficient conditions to ensure it. Some of the most commonly used conditions are:

- **Condition 1:**  $|Z_{T \wedge n}| \leq M$  for all  $n$

- **Condition 2:**  $E[Z_{T \wedge n}^2] \leq M$  for some constant  $M$

- **Condition 3:**

- (i)  $E[T] < \infty$ ;

- (ii) There exists a constant  $K$  such that  $E[|Z_{n+1} - Z_n| | X_1, \dots, X_n] < K$ , for  $n < T$

The interchange conditions are very crucial for the martingale stopping theorem to hold!

## Example: Gambler's Ruin Problem Revisited

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- Consider an individual who starts at position  $k$  and at each step either moves 1 position to the right with probability 0.5 or 1 position to the left with probability 0.5. Assume that the successive movements are independent.
- Suppose that he will stop moving when he reaches either  $N$  or 0. What is the probability that he stops at  $N$ ?

$$X_i = \begin{cases} 1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}, \quad Z_n = \sum_{i=1}^n X_i$$

Define  $T = \{\min n : \text{reach } N \text{ or } 0\}$

$$E[Z_T] = \overset{k}{E[Z_0]} \quad (\text{martingale stopping theorem})$$

$$Z_\infty = \begin{cases} 0 & ; q = 1-p \\ N & ; p \end{cases}$$

$$\Rightarrow E[Z_T] = Np + 0 \cdot q = k$$

$$\boxed{\therefore p = \frac{k}{N}} \quad \#$$

## Example: Expected Number of Steps

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- Consider the preceding example. What is the expected number of steps this individual takes to reach  $N$  or  $0$ ?

Define  $Y_n = \left(\sum_{i=1}^n X_i\right)^2 - n \text{Var}[X_i]$  (variance process  $\rightarrow$  Martingale).

$$E[Y_T] = E[Y_0] = k^2$$

$$E[Y_T] = E[Z_T^2 - T]$$

$$= E[Z_T^2] - E[T]$$

$$= N^2 \cdot P(Z_T = N) + 0^2 \cdot P(Z_T = 0) - E[T]$$

$$= N^2 \left(\frac{k}{N}\right) + 0 - E[T]$$

$$= kN - E[T]$$

$$\therefore kN - E[T] = k^2$$

$$\therefore E[T] = k(N - k) \neq$$

$$Z_T = \begin{cases} 0 & ; 1 - \frac{k}{N} \\ N & ; \frac{k}{N} \end{cases}$$

## Example: Unfair Games

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- Now suppose that the individual moves to his right with a higher probability. That is, let

$$P(X_i = 1) = p, \quad P(X_i = -1) = q = 1 - p,$$

with  $p > q$ .

- Let

$$S_n = \sum_{i=1}^n X_i.$$

- Verify that  $Z_n = (q/p)^{S_n}$  is a martingale. Using it and the martingale stopping theorem, we can compute the probability that he reaches  $N$  earlier than 0.

Show that  $E[z_{n+1} | x_1, \dots, x_n] = z_n$ ,

$$E[z_{n+1} | x_1, \dots, x_n] = E\left[\left(\frac{q}{p}\right)^{\sum_{i=1}^{n+1} x_i} \mid x_1, \dots, x_n\right]$$

$$= \left(\frac{q}{p}\right)^{s_n} E\left[\left(\frac{q}{p}\right)^{x_{n+1}}\right]$$

$$E\left[\left(\frac{q}{p}\right)^{x_{n+1}}\right] = \left(\frac{q}{p}\right)^1 \cdot p + \left(\frac{q}{p}\right)^{-1} \cdot q = 1$$

$$\therefore E[z_{n+1} | x_1, \dots, x_n] = \left(\frac{q}{p}\right)^{s_n} = z_n \quad \#$$

Martingale stopping theorem:  $E[z_T] = E[z_0] = \left(\frac{q}{p}\right)^k$

By Definition:  $E[z_T] = \left(\frac{q}{p}\right)^N \cdot p(z_T = N) + \left(\frac{q}{p}\right)^0 \cdot [1 - p(z_T = N)]$

↓ solve for  $p(z_T = N)$  #

## Financial Application

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Consider a stock price  $S_n$  that is a martingale under the risk-neutral measure. Let  $\tau$  be the first time  $S_n$  hits a certain barrier

**Martingale Stopping Theorem** allows us to compute the expected payoff:

$$\mathbb{E}[f(S_\tau)] = f(S_0) \quad (\text{under certain conditions})$$



## Martingale Inequalities

# Markov Inequality & Chebyshev Inequality

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**Markov Inequality.** For a non-negative random variable  $X$ , for any  $a > 0$

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

**Chebyshev Inequality:** For any random variable  $X$  with mean  $\mu$  and variance  $\sigma^2$ ,

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2},$$

# Weak Law of Large Number (WLLN)

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## Theorem

Let  $X_1, X_2, \dots$  be i.i.d. random variables with  $\mathbb{E}[X_1] = \mu$  and  $\text{Var}[X_1] = \sigma^2$ . Define  $S_n = \sum_{i=1}^n X_i$ . Then for any  $\varepsilon > 0$ :

$$\lim_{n \rightarrow \infty} P \left( \left| \frac{S_n}{n} - \mu \right| \geq \varepsilon \right) = 0.$$

Equivalently:

$$\frac{S_n}{n} \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty.$$

# Doob's Maximal Inequality

## Theorem (Doob's Maximal Inequality)

Let  $(M_n)_{n \geq 0}$  be a non-negative martingale. Then for any  $a > 0$ :

$$\mathbb{P} \left( \max_{0 \leq k \leq n} M_k \geq a \right) \leq \frac{\mathbb{E}[M_n]}{a}$$

### Intuition:

- The probability that the maximum up to time  $n$  exceeds  $a$  is bounded by the expected value at time  $n$  divided by  $a$ .
- Similar to Markov's inequality, but for the maximum rather than just the final value.

## Example: Stock Price Martingale

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Suppose  $M_n$  is a non-negative martingale representing a stock price. Let  $\mathbb{E}[M_n] = S$  be the expected price at time  $n$ . What is the probability that the stock price ever doubles  $S$ ?

$$P(\max_{1 \leq k \leq n} M_k > 2\mathbb{E}[M_n]) = \frac{\mathbb{E}[M_n]}{2\mathbb{E}[M_n]} = \frac{1}{2} \neq$$

# Kolmogorov's Inequality

## Theorem (Kolmogorov's Inequality)

Let  $X_1, \dots, X_n$  be independent random variables with  $\mathbb{E}[X_i] = 0$  and  $\text{Var}(X_i) < \infty$ .

Define  $S_k = X_1 + \dots + X_k$ . Then for any  $\lambda > 0$ :

$$\mathbb{P} \left( \max_{1 \leq k \leq n} |S_k| \geq \lambda \right) \leq \frac{\text{Var}(S_n)}{\lambda^2}$$

## Comparison with Chebyshev:

- Chebyshev:  $\mathbb{P}(|S_n| \geq \lambda) \leq \frac{\text{Var}(S_n)}{\lambda^2}$
- Kolmogorov: Same bound but for the **maximum** over the entire path
- Much stronger!

## Example: Random Walk

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Consider a simple symmetric random walk with  $S_n = \sum_{i=1}^n \pm 1$ , starting at 0. What is  $\mathbb{P}(\max_{1 \leq k \leq 100} |S_k| \geq 20)$ ?

$$\mathbb{P}(\max_{1 \leq k \leq 100} |S_k| \geq 20) \leq \frac{\text{Var}[S_{100}]}{400} = \frac{100 \cdot \text{Var}[X_i]}{400} = \frac{100(1)}{400} = \frac{1}{4} \neq$$

# Strong Law of Large Numbers (SLLN)

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## Theorem

Let  $X_1, X_2, \dots$  be i.i.d. random variables with  $\mathbb{E}[X_1] = \mu$  and  $\text{Var}[X_1] = \sigma^2$ . Define  $S_n = \sum_{i=1}^n X_i$ . Then:

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1.$$

Equivalently:

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \rightarrow \infty.$$



# Borel–Cantelli

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Let  $A_1, A_2, \dots$  be events. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$$

then

$$\mathbb{P}(A_n \text{ occurs infinitely often}) = 0$$

$$P\left(\max_{1 \leq k \leq n} \left| \frac{S_k}{n} \right| \geq \varepsilon\right) = P\left(\max_{1 \leq k \leq n} |S_k| \geq n\varepsilon\right) = \frac{\text{Var}(S_n)}{n^2 \varepsilon^2}$$

$$= \frac{n\sigma^2}{n^2 \varepsilon^2} = \frac{\sigma^2}{n \varepsilon^2}$$

Apply Borel-Cantelli:

$$\sum_{n=1}^{\infty} P\left(\max_{1 \leq k \leq n} |S_k| \geq n\varepsilon\right) = \sum_{n=1}^{\infty} \frac{\sigma^2}{n \varepsilon^2} \rightarrow \infty \quad (\text{Can't be used})$$

Instead apply to  $m^2$  rather than  $n$ .

$$\therefore P\left(\max_{1 \leq k \leq m^2} |S_k| \geq m^2 \varepsilon\right) = \frac{\sigma^2}{m^2 \varepsilon^2}$$

$$\therefore \sum_{m=1}^{\infty} P\left(\max_{1 \leq k \leq m^2} |S_k| \geq m^2 \varepsilon\right) = \sum_{m=1}^{\infty} \frac{\sigma^2}{m^2 \varepsilon^2} < \infty \quad \text{as} \quad \sum_{m=1}^{\infty} \frac{1}{m^2} < \infty$$

Then, by Borel-Cantelli,

$$P(\max_{1 \leq k \leq m^2} |S_k| \geq m^2 \varepsilon \text{ occurs infinitely often}) = 0. \quad \#$$

For sufficiently large  $m$ ,  $P\left(\frac{\max_{1 \leq k \leq m^2} |S_k|}{m^2} > \varepsilon\right) = 0.$

$$\begin{aligned} \text{For } n \in [m^2, (m+1)^2], \quad \frac{\max_{1 \leq k \leq n} |S_k|}{n} &\leq \frac{\max_{1 \leq k \leq (m+1)^2} |S_k|}{m^2} \\ &= \frac{\max_{1 \leq k \leq (m+1)^2} |S_k|}{(m+1)^2} \cdot \frac{(m+1)^2}{m^2} \rightarrow 0 \end{aligned}$$

a.s.  
#

Therefore,  $\frac{\max_{1 \leq k \leq m^2} |S_k|}{m^2} \rightarrow 0$  a.s. #